Automated Defect Detection in Electroluminescence Imaging of PV Cells Using Deep Learning

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Problem

- Energy Consumption and Key Sectors Electrification
- Renewable Energy and Solar Power Generation Investment
- Invisible Defects and Photovoltaic Efficiency
- Data Storage and Data Processing Bottleneck

Solution

- Remote Inspections and Data Collection Procedures
- Thermography Analysis and Infra-Red Sensors Limitations
- Electroluminescence Imaging and Predictive Advantages
- Intelligent Data Analysis and Deep Learning Models

Dataset [1]

Quantity:	2,624 Images » 300 x 300 x 1 » 8-bit
Annotations:	Image Level Pseudo-Labels
4 Classes:	0.00 » 0.33 » 0.66 » 1.00
Binary Classification:	15% Data Discarded
Distribution:	901 MonoCrystalline » 1322 PolyCrystalline
Class Imbalance:	65 - 70% Functional » 35 - 30% Defective
Dataset Split:	80% Train » 10% Validation » 10% Test

Architecture

CNN VGG19 [2]: 2 Blocks » 19 Layers » 33,133,121 Parameters

Convolutional: 5 Blocks » 16 Layers » 20,024,384 Parameters

Fully Connected: 1 Block » 3 Layers » 13,108,737 Parameters

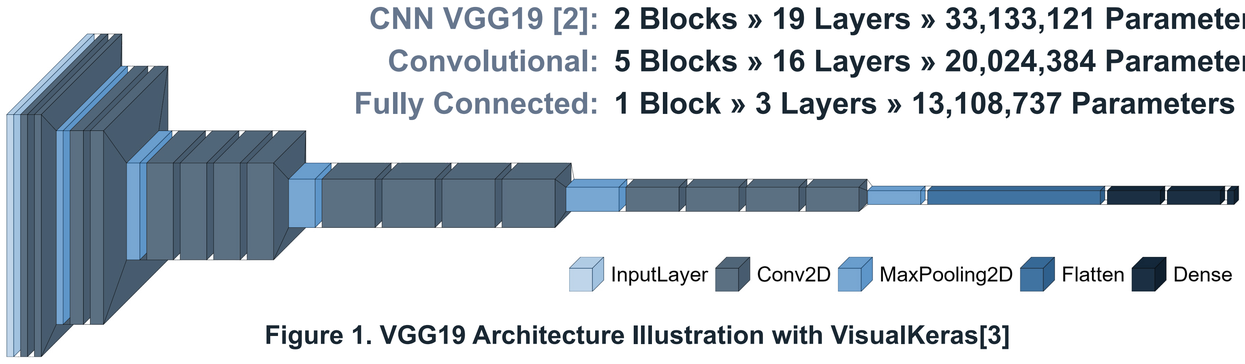


Figure 1. VGG19 Architecture Illustration with VisualKeras[3]

Training

Augmentation:	On-the-Fly » Pre-Processing Layers
Transfer Learning:	ImageNet Pre-Trained Weights
Fine-Tuning:	Conv2D Layers Initially Frozen
Regularization:	DropOut and Loss Penalty

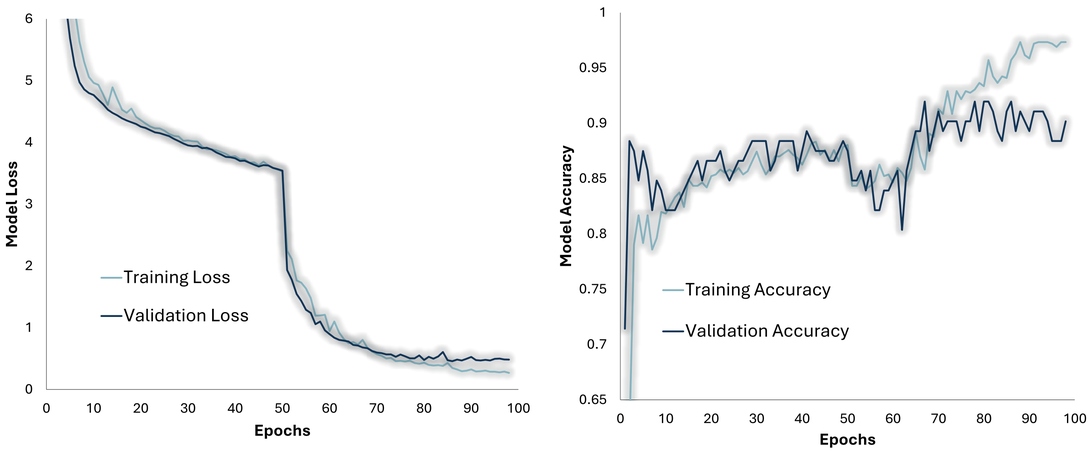


Figure 2. Loss and Accuracy Evolution during Model Training

Results

Table 1. PV Cells Test inference Results Comparison to Reference Work [4]

Results	AUC			TP rate			TN rate		
	mono	poly	mix	mono	poly	mix	mono	poly	mix
Reference	95,2%	93,8%	94,4%	90,3%	86,1%	87,9%	82,8%	86,0%	84,7%
Presented	94%	95%	90%	93,2%	93,9%	99,5%	84,6%	84,0%	77,9%

Evaluation

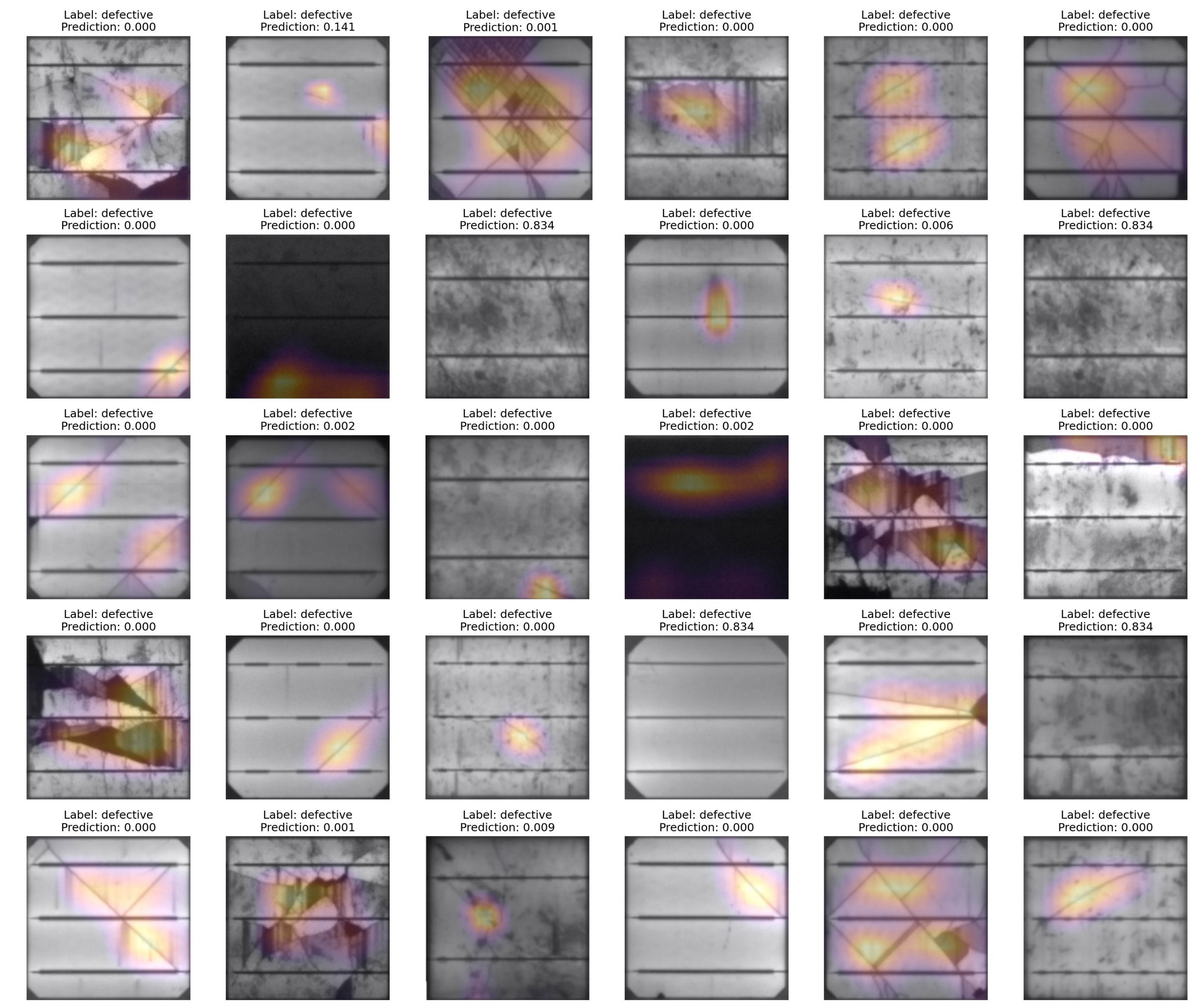


Figure 3. Defective PV Cells Test Results and Heatmap using GradCAM++ [5]

Conclusion

- Deep Learning VS Traditional Computer Vision
- Inspection System Modeling Efficiency
- Model Explainability and Testing Extension
- Dataset Quality and Annotations Noise

Future

- Power Efficient Measurement as Labels
- Time-Series Data for Predictive Maintenance

References

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